65. Suppose the average weight of 9 persons is 50 kg. The average weight of the first 5 persons is 45 kg, whereas the average weight of the last 5 persons is 55 kg. Then the weight of the 5th person will be

(a) 45 kg (b) 47.5 kg
(c) 50 kg (d) 52.5 kg

Ans. (c) :
Total weight of 9 persons = 50 × 9 = 450 kg
Total weight of the first 5 person = 45 × 5 = 225 kg
and total weight of the last 5 person = 55 × 5 = 275 kg
= (225 + 275) – 450
= 500 – 450 = 50 kg

66. In a city, 12% of households earn less than Rs. 30,000 per year, 6% households earn more than Rs. 2,00,000 per year, 22% households earn more than Rs. 1,00,000 per year and 990 households earn between Rs. 30,000 and Rs. 1,00,000 per year. How many households earn between Rs. 1,00,000 and Rs. 2,00,000 per year?

(a) 250 (b) 240
(c) 230 (d) 225

Ans. (b) :
6% > 20,0000……..(i)
22% > 100000……(ii)
12% < 30,000……..(iii)
from eq^{n} (i) & (ii)
20,0000 > (20 – 6)% >10,0000
20,0000 > 16% > 10,000
The percentage between 30,000 and 10,0000 households=(100-(16+12+6)]%
→(100–34)%
=66%
∴ 66%=990 households
$\therefore 100\% = \frac{990}{66} \times 100$ households ⇒ 1500
The number of households earn between 100000 to

$$20000 \Rightarrow 1500 \times \frac{16}{100}$$
$$= 240$$

67. There are thirteen 2-digit consecutive odd numbers. If 39 is the mean of the first five such numbers, then what is the mean of all the thirteen numbers?

(a) 47 (b) 49
(c) 51 (d) 45

Ans. (a) :
Let the first number =x
(a)

$$\frac{x+(x+2)+(x+4)+(x+6)+(x+8)}{5} = 39$$

then, $\frac{5x+20}{5} = 39$

$x + 4 = 39$

$\boxed{x = 35}$

then, Series⇒ 35, 37, 39 ……13th

$$S_n \frac{n}{2}[2a+(n-1)d]$$

$$S_{13} = \frac{13}{2}[2\times 35+(13-1)2]$$

$$S_{13} = \frac{13}{2}[70+24]$$

$$S_n = \text{Required Average} = \frac{\frac{13}{2}(70+24)}{13}$$

$$= \frac{1}{2}[94]$$

$$= 47$$

68. There is a milk sample with 50% water in it. If 1/3rd of this milk is added to equal amount of pure milk, then water in the new mixture will fall down to

(a) 25% (b) 30%
(c) 35% (d) 40%

Ans. (a) :
Let total amount of mixture = x

$\frac{1}{3}$rd of mixture = $\frac{1}{3}$ x

⇒	Water	Milk
	$\frac{1}{3}\times x\frac{1}{2}$	$\frac{1}{3}\times x\frac{1}{2}$
⇒	$\frac{x}{6}$	$\frac{x}{6}$

∵ $\frac{x}{3}$ pure milk added to mixture,

Total amount of mixture = $\frac{x}{3}+\frac{x}{3}=\frac{2x}{3}$

Hence, The percentage of water in new

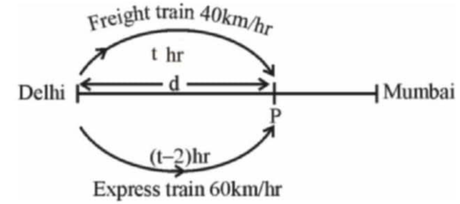
$$\text{mixture} = \frac{\frac{x}{6}}{\frac{2x}{3}} \times 100\%$$

$$\Rightarrow \frac{x}{6}\times\frac{3}{2x}\times 100\% = 25\%$$

69. A freight train left Delhi for Mumbai at an average speed of 40 km/hr. Two hours later, an express train left Delhi for Mumbai, following the freight train on a parallel track at an average speed of 60 km/hr. How far from Delhi would the express train meet the freight train?

(a) 480 km (b) 260 km
(c) 240 km (d) 120 km

Ans. (c) :



∵ distance =speed × time
∴ d= 40 ×t ---------- (i)
d= 60 × (t – 2) ------ (ii)
From eg (i) & (ii)
40t = 60 (t –2)
40t =60 60t –120
$\boxed{t = 6hr}$

∴ Required distance ⇒ d =40 ×t
d =40 ×6
d =240/cm

70. Certain 3-digit numbers following characteristics :

1. All the three digits are different.
2. The number is divisible by 7.
3. The number on reversing the digits is also divisible by 7.

How many such 3-digit numbers are there?

(a) 2 (b) 4 (c) 6 (d) 8

Ans. (b) The Numbers are :
168
861
259
952
So that total Numbers (4).

71. How many numbers are there between 99 and 1000 such that the digit 8 occupies the units place?

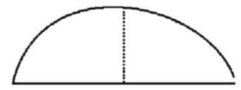
(a) 64 (b) 80 (c) 90 (d) 104

Ans. (c) : First number = a ⇒108
and last number =*l* ⇒998
∴ difference d ⇒ 10
∵ In = a + (n –1) d
l = a + (n – 1) d
998 =108 + (n – 1)10
998 –108 = (n – 1)10
$\frac{990}{10} - n - 1$
89 – n – 1
n = 89 + 1
n = 90

72. If for a sample data
Mean < Median < Mode
Then, the distribution is

(a) symmetric
(b) skewed to the right
(c) neither symmetric nor skewed
(d) skewed to the left

Ans. (d) :



If for sample data
mean < median <mode
then the distribution skewed to the left
and if for a sample date
mean > median > mode
then the distribution is skewed to the right

73. The age of Mr. X last year was the square of a number and it would be the cube of a number next year. What is the least number of years he must wait for his age to become the cube of a number again?

(a) 43 (b) 38 (c) 25 (d) 16

Ans. (b) :
According to question,
$A^3 – B^2 = 2$
If, A = 3, B = 5
$(3)^3 – (5)^2 = 2$
27 – 25 = 2
2 = 2
L.H.S. =R.H.S.
Hence,
Present age = 26
∵ $(3)^3 = 27$
and $(4)^3 = 64$
Required year for waiting = 64 – 26
=38 years

74. P works thrice as fast as Q, whereas P and Q together can work four times as fast as R. If P, Q and R together work on a job, in what ratio should they share the earnings?

(a) 3 : 1 : 1 (b) 3 : 2 : 4
(c) 4 : 3 : 4 (d) 3 : 1 : 4

Ans. (a) :
P : Q = 3 : 1 _____ (i)
(a) and (P + Q) : R = 4 : 1____ (ii)
From eq (i) & (ii),
∴ P : Q : R = 3 : 1 : 1